

Echolalia in Post-Stroke Aphasia: Prevalence and Linguistic Correlates

Florencia Hoet¹, Diana López-Barroso^{1,2,3}, Sergio Vertedor¹, Africa Gómez-Pérez^{1,2,3}, Guadalupe Dávila^{1,2,3},
Marcelo L. Berthier^{1,2}, María José Torres-Prioris^{*1,2,3}

¹ Cognitive Neuroscience and Aphasia Unit, Centro de Investigaciones Médico-Sanitarias, University of Malaga, Malaga, Spain; ² Instituto de Investigación Biomédica de Málaga - IBIMA, Malaga, Spain; ³Area of Psychobiology, Faculty of Psychology and Speech Therapy, University of Malaga, Malaga, Spain

Background

Echolalia, the repetition of previously heard utterances, is a classical symptom of aphasia, described for over a century. It ranges from automatic (verbatim) to mitigated (modified) repetition (Pick, 1902; Stengel, 1947). Traditionally, echolalia has been associated with transcortical aphasias (Geschwind et al., 1968; Berthier, 1999; López-Barroso et al., 2023).

Despite its clinical relevance, echolalia has never been systematically investigated in large cohorts of individuals with post-stroke aphasia. Consequently, its prevalence, distribution across aphasia subtypes, and relationships with aphasia severity and core language processes remain poorly understood.

Aims:

- Examine echolalic productions in persons with post-stroke aphasia (PSA) compared to healthy controls.
- Characterize echolalia patterns across aphasia subtypes and task contexts.
- Identify independent linguistic predictors of automatic and mitigated echolalia.

Methods

Participants. 128 participants: 61 people with PSA (M age = 54.72 years; 22 women) and 67 matched healthy controls (M age = 54.82 years; 30 women).

Instruments. Language was assessed using the Western Aphasia Battery–Revised (WAB-R). Echolalic productions were coded across the six auditory subtests and classified as automatic or mitigated echolalia.

Statistical Analysis.

- Group differences: Mann–Whitney U tests with Bonferroni correction.
- Subtype and severity: Kruskal–Wallis tests, Dunn post-hoc comparisons.
- Predictors: robust regression (MM-estimation); tercile analyses for auditory comprehension.

Results

- Group differences.** Participants with PSA showed more total, automatic, and mitigated echolalia than controls (all $p < .001$; *Figure 1*).
- Prevalence & subtypes.** Echolalia occurred in 91% of PSA participants and across all aphasia subtypes, with the majority of individuals in each aphasia subtype producing at least one echolalic response (range: 89.7–100% across subtypes).
- Severity.** Automatic echolalia correlated with aphasia severity ($r = -.39$, $p = .002$; *Figure 2*), whereas mitigated echolalia did not ($r = -.13$, $p = .327$).
- Task context.** Echolalia occurrence varied significantly across tasks ($p < .001$), with the highest rates observed in auditory word recognition and the lowest in conversation (*Figure 3*).
- Linguistic predictors.** Automatic echolalia was predicted exclusively by auditory comprehension ($\beta = -2.74$, $p = .003$), whereas mitigated echolalia was predicted by both comprehension ($\beta = -7.79$, $p < .001$) and repetition ability ($\beta = 1.40$, $p = .048$).

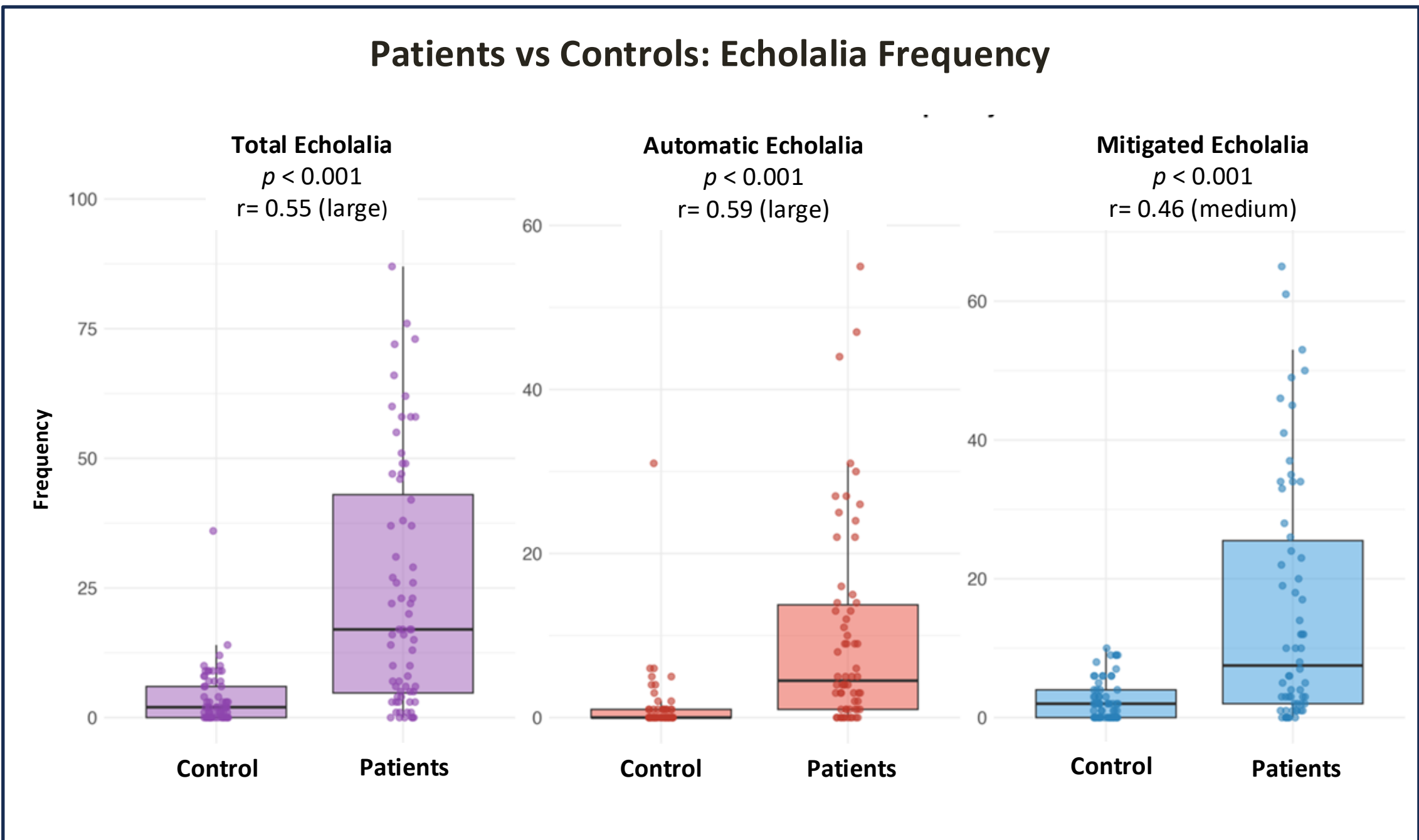


Figure 1

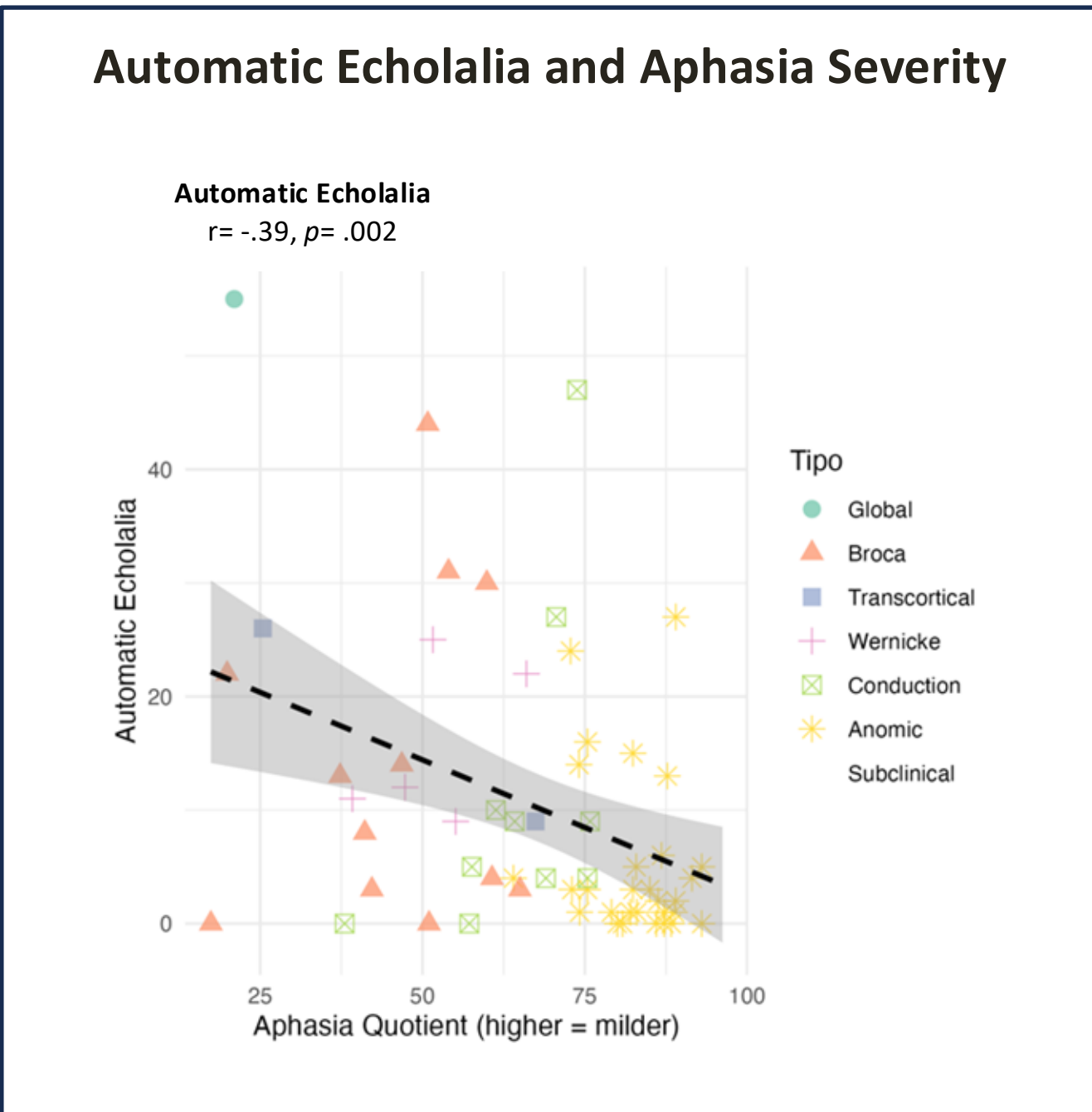


Figure 2

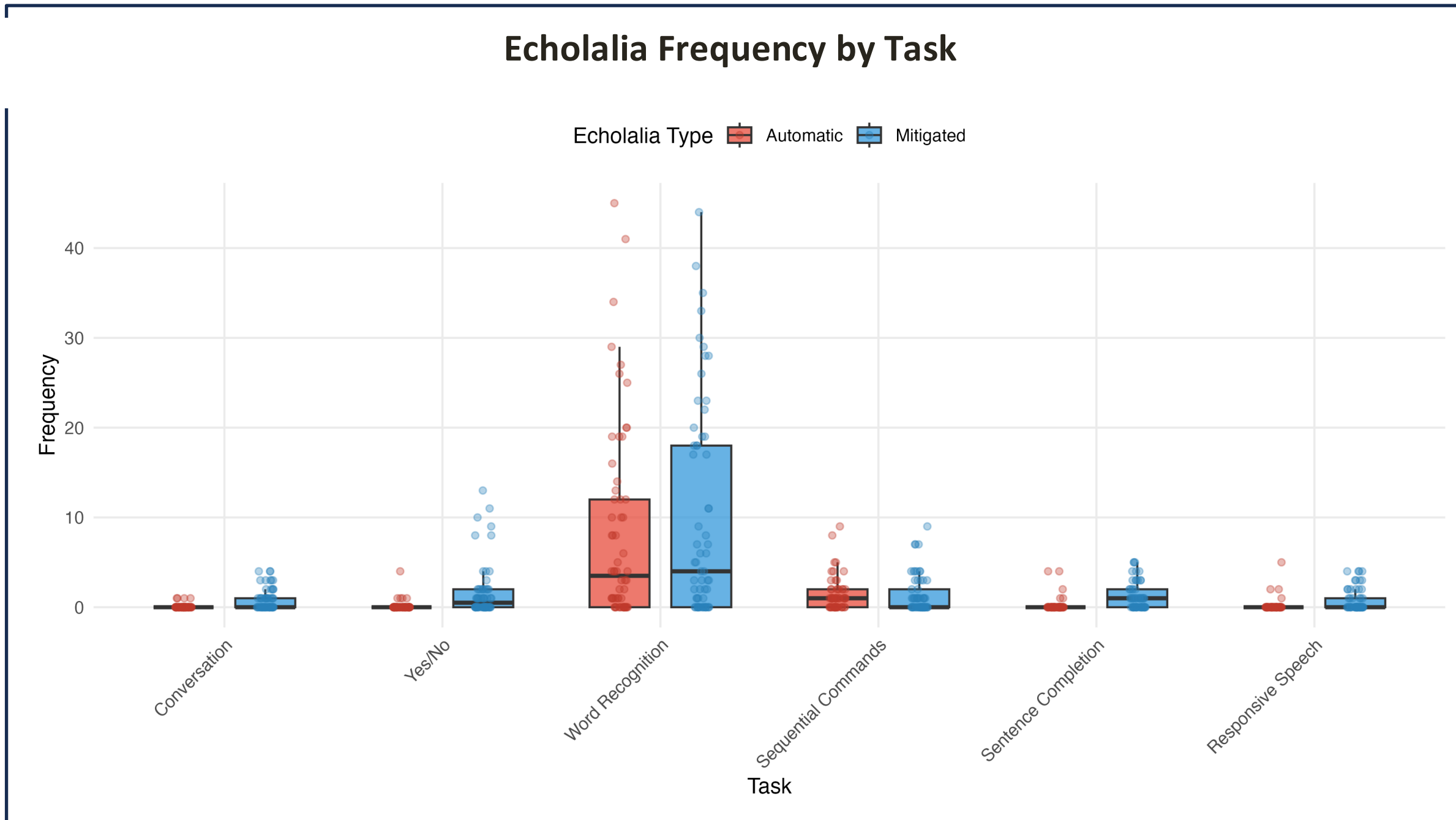


Figure 3

Conclusions

- Echolalia is highly prevalent in PSA and occurs across all aphasia subtypes, not only in transcortical syndromes.
- Aphasia severity and auditory comprehension, rather than categorical subtype, primarily explain the frequency of echolalia.
- Automatic echolalia appears to reflect overall language impairment, whereas mitigated echolalia additionally depends on repetition abilities.
- Task structure strongly modulates echolalic production, with recognition-based tasks eliciting the highest frequencies.

References

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UNIVERSIDAD
DE MÁLAGA



Contact: mjprioris@uma.es